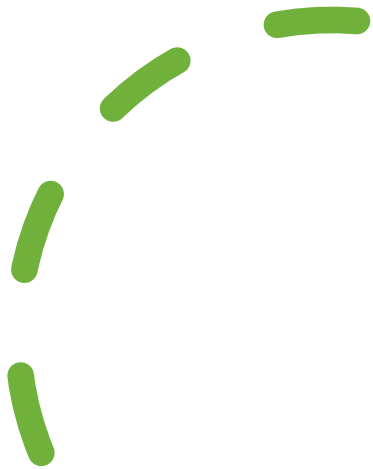


Decision Trees



Agenda

- Assumptions
- Introduction
- The basics: What, Why and How?
- Understanding the concept
- Practical Implementation



Introduction

Bird or animal?

How do you identify so quickly?
You know the rules!



How to teach a computer?

- How do you teach the same thing to a computer?
- Same thing = “Set of Rules”
- Decision Trees!

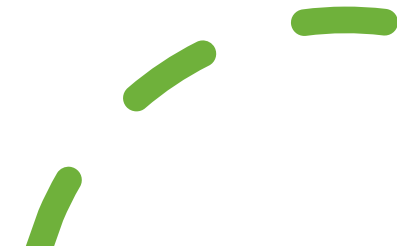
What is it?

- A tree that helps you take a decision! Really.
- Fun fact – it's an inverted tree*
- We create it daily – thousands of times!



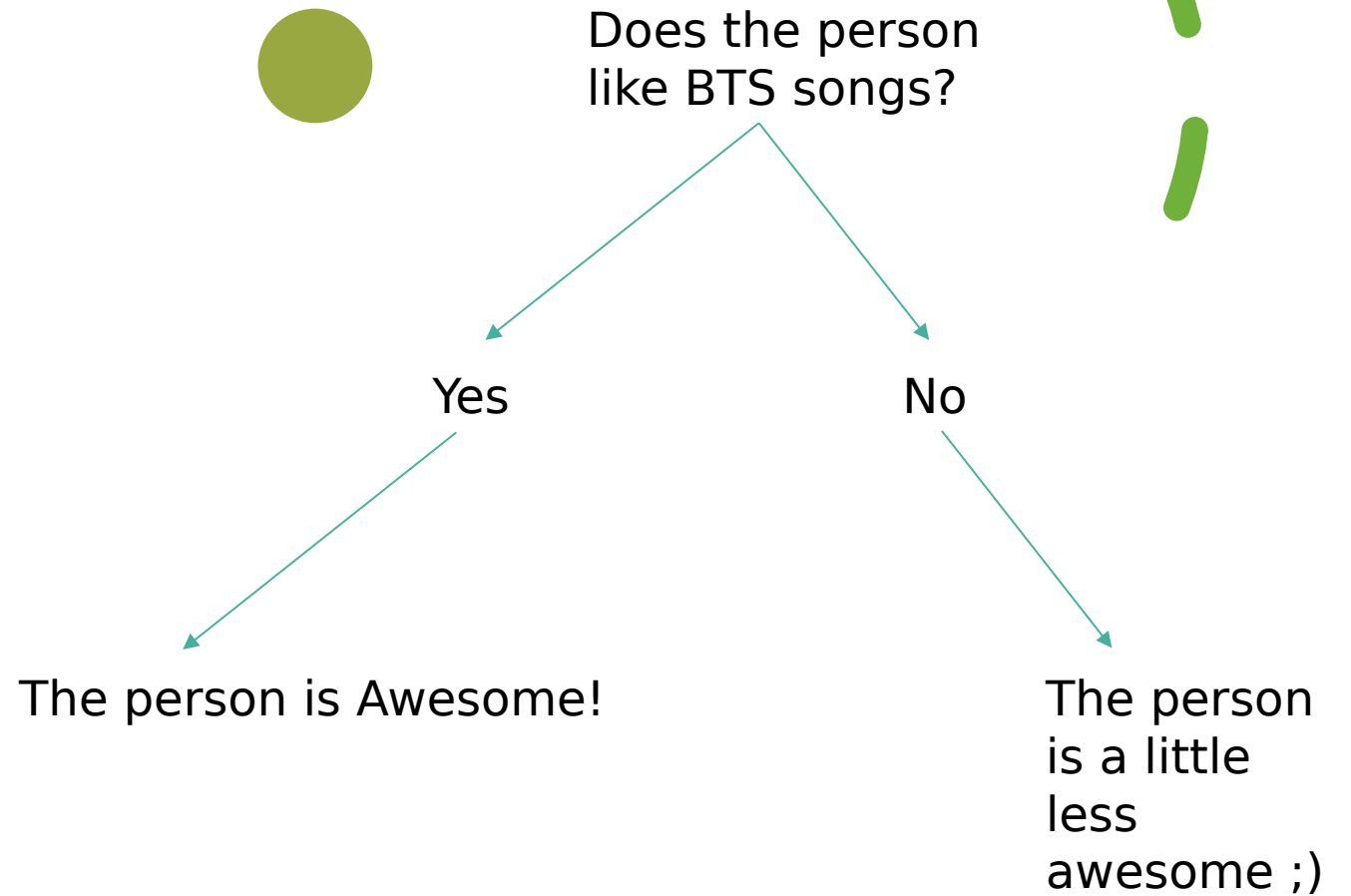


What is it?

- Have you driven? Have you cooked? Have you thought?
 - Anticipation / What – If?
 - In milliseconds
 - Let's meet my friend Bob.
- 

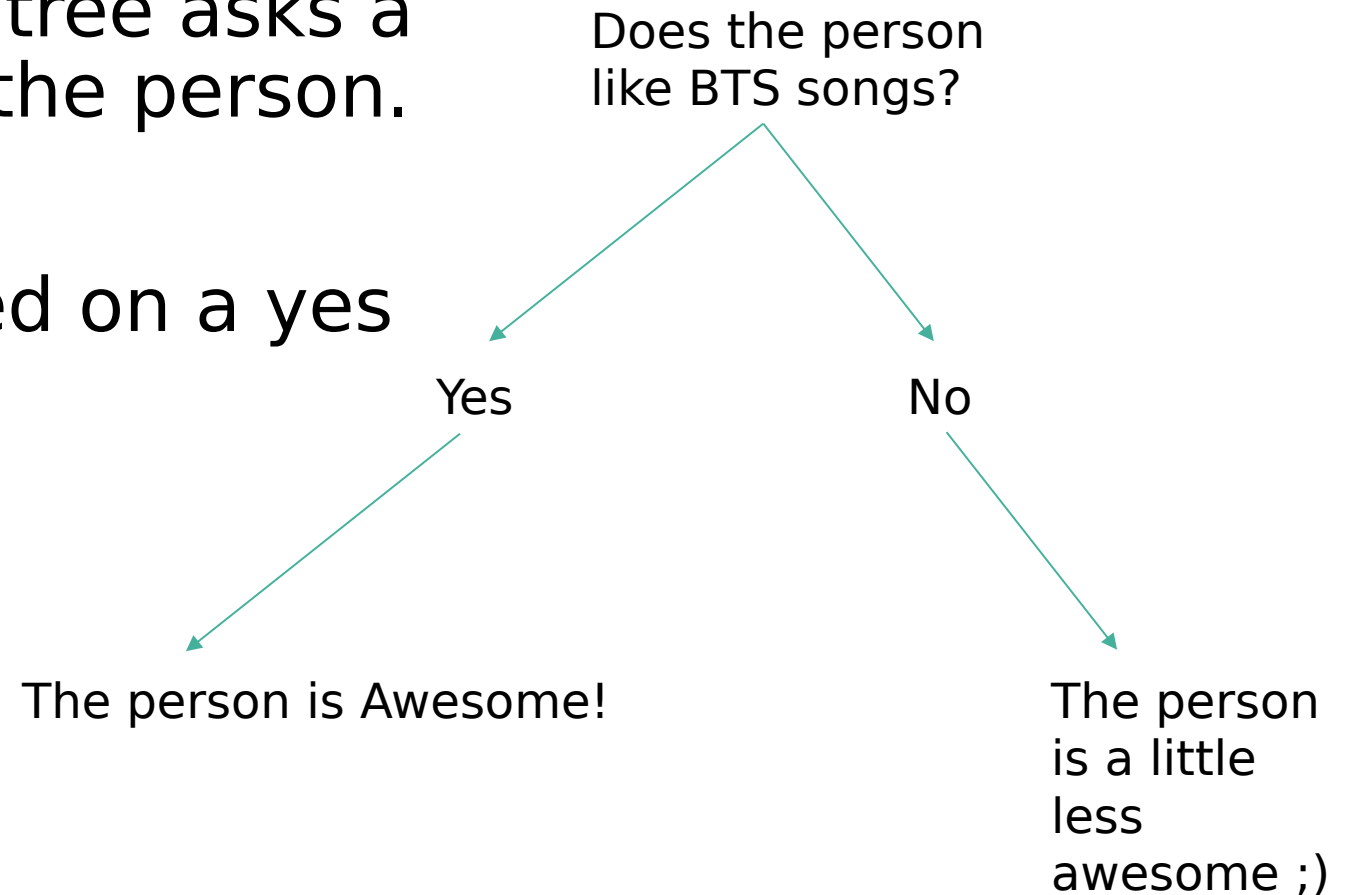
What is it?

- On a serious note
- Let's simplify



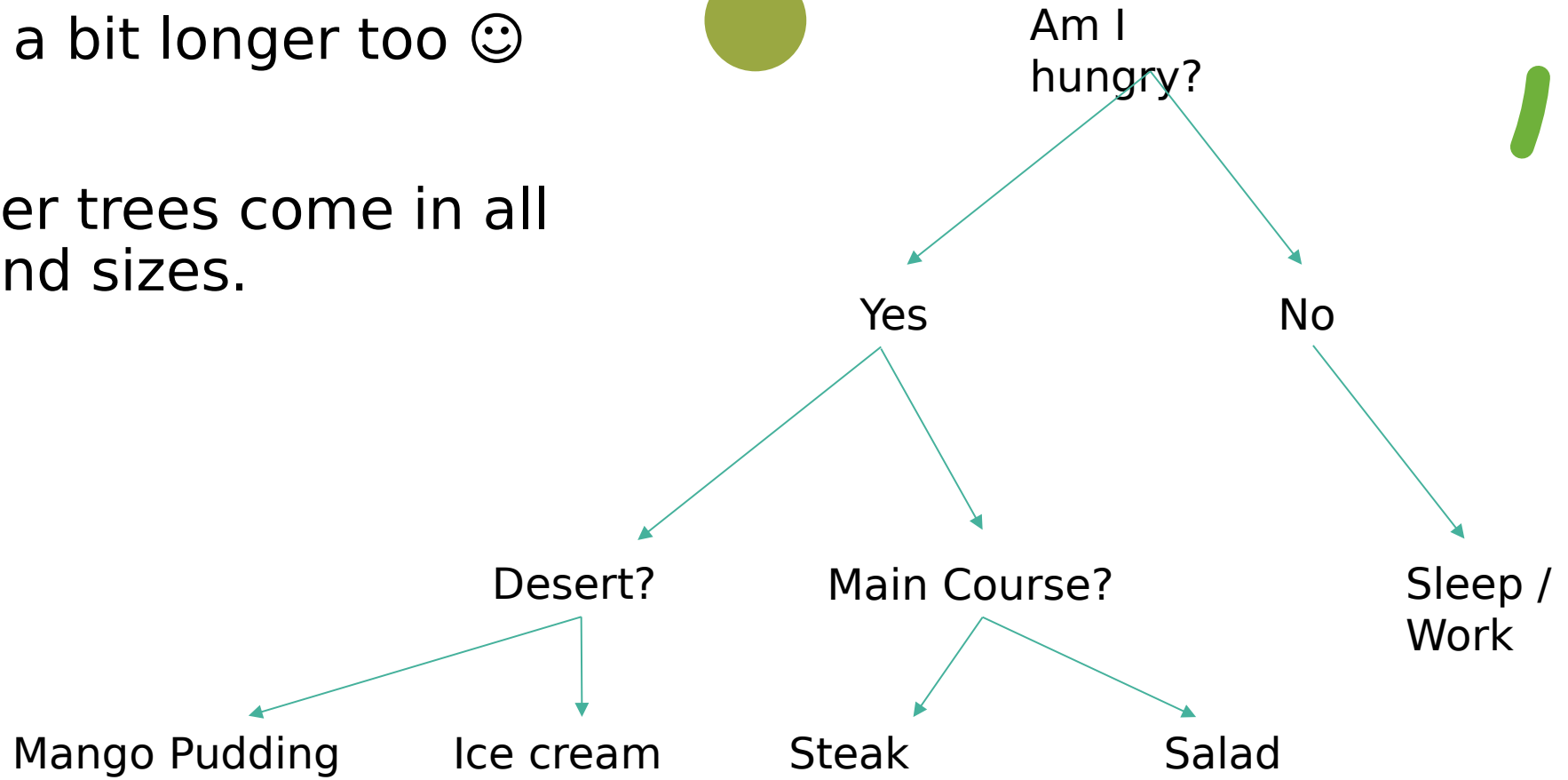
What is it?

- So essentially a decision tree asks a question and **classifies** the person.
- This decision tree is based on a yes / no question.



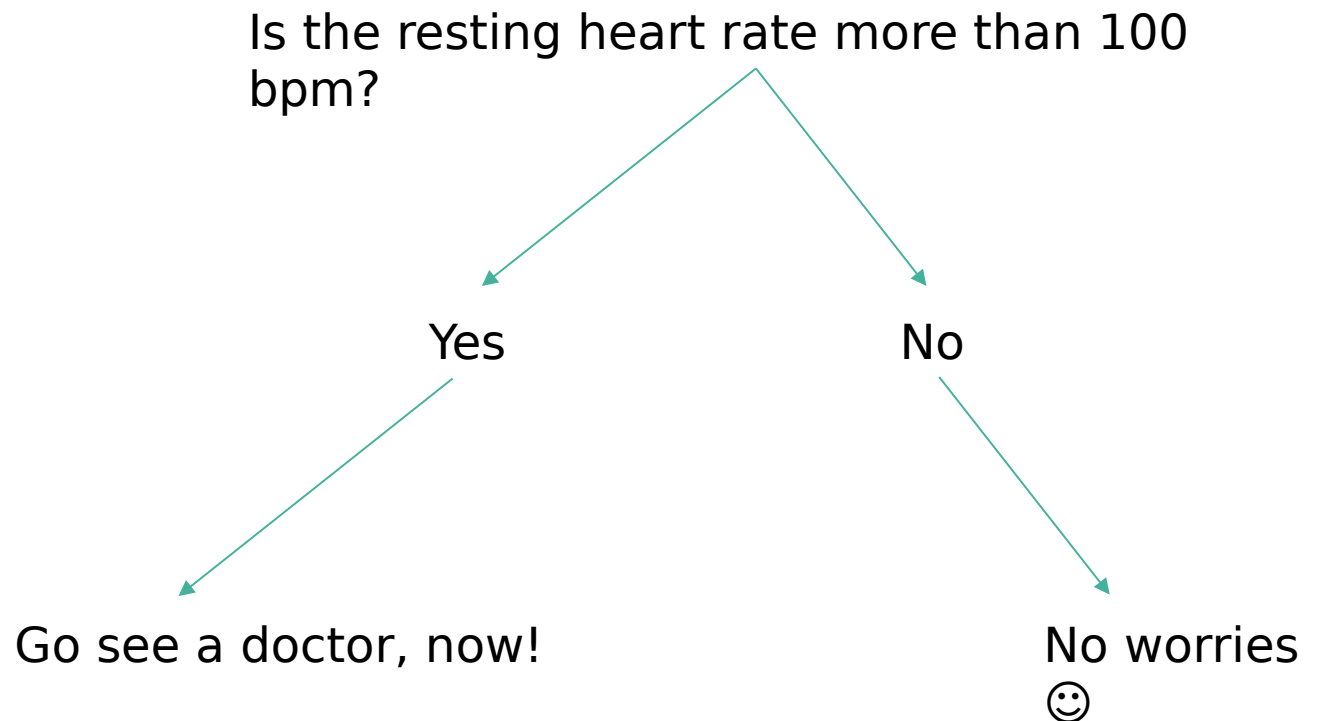
What is it?

- It can be a bit longer too 😊
- Remember trees come in all shapes and sizes.



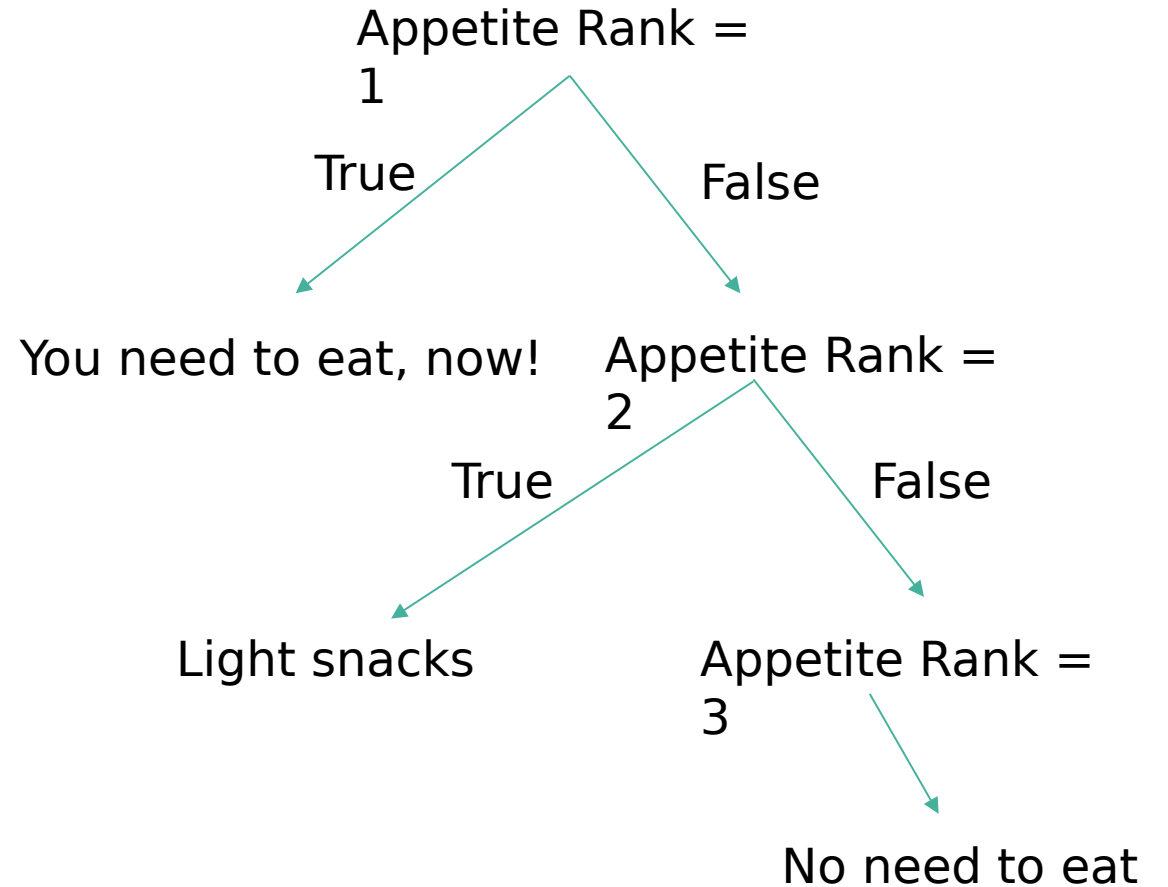
What is it?

- But wait, you can do this for numeric data too?



What is it?

- How about ranked data?
- How hungry are you?
- 1 – Super Hungry
- 2 – Moderate
- 3 – Not hungry



What is it?

- The classification can be categories or numeric – both.

Is the resting heart rate more than 100 bpm?

Yes

No

Go see a doctor, now!

No worries



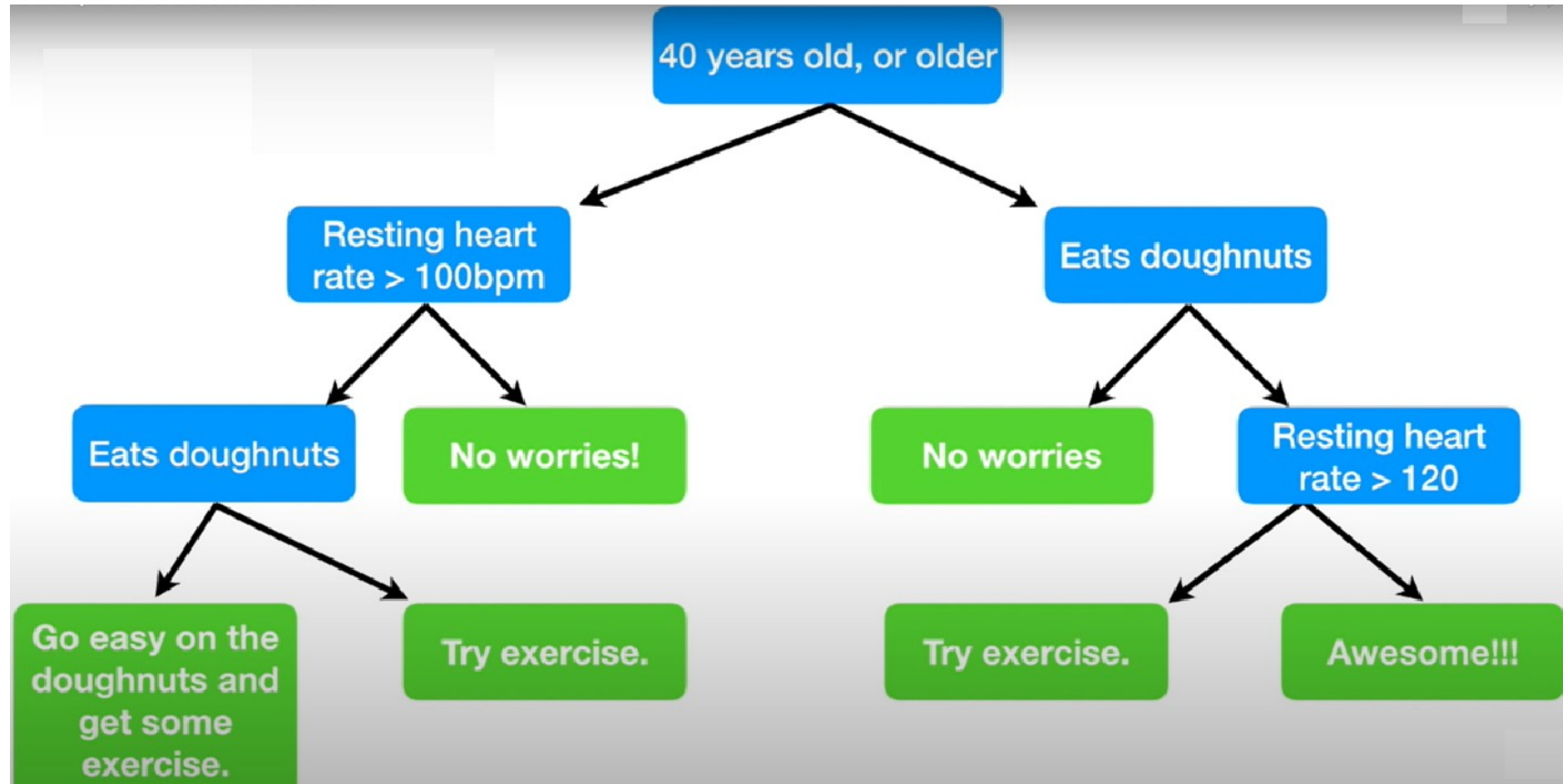
Mouse weighs between 15 – 20 grams

It is less
than
180mm
long

It is more
than
180mm
long

What is it?

- It can be a mix of all these too!

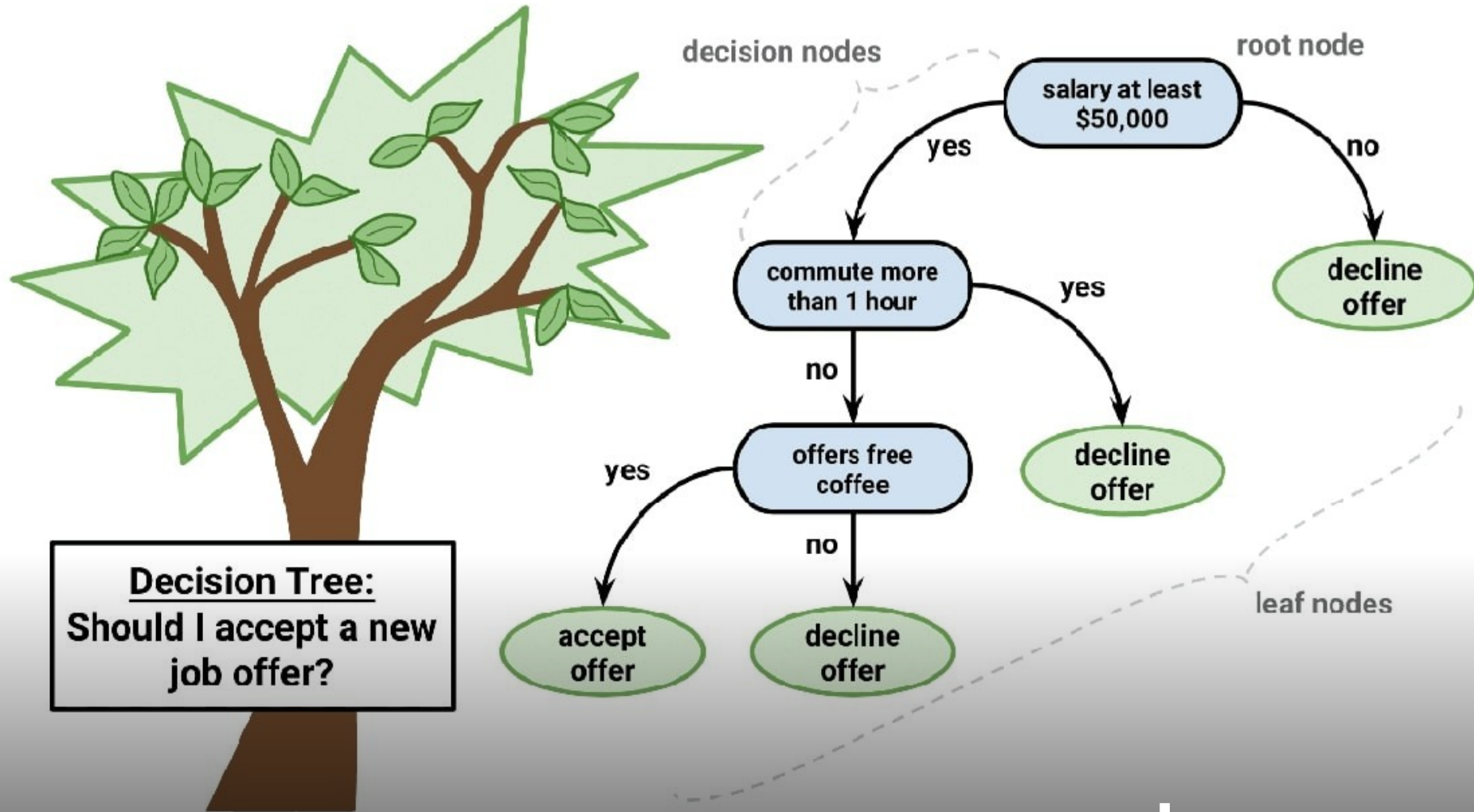


To summarize

- You start at the top
- Keep going down
- Until you can't anymore

Why are we interested in it?

- It's visual and therefore easy
- Helps businesses relate and validate the rules / conditions
- Intuitive
- Works well with numbers and categories both
- Less data cleaning required
- Non Parametric method. This means that decision trees have no assumptions about the space distribution and the classifier structure.



Jargons

- 

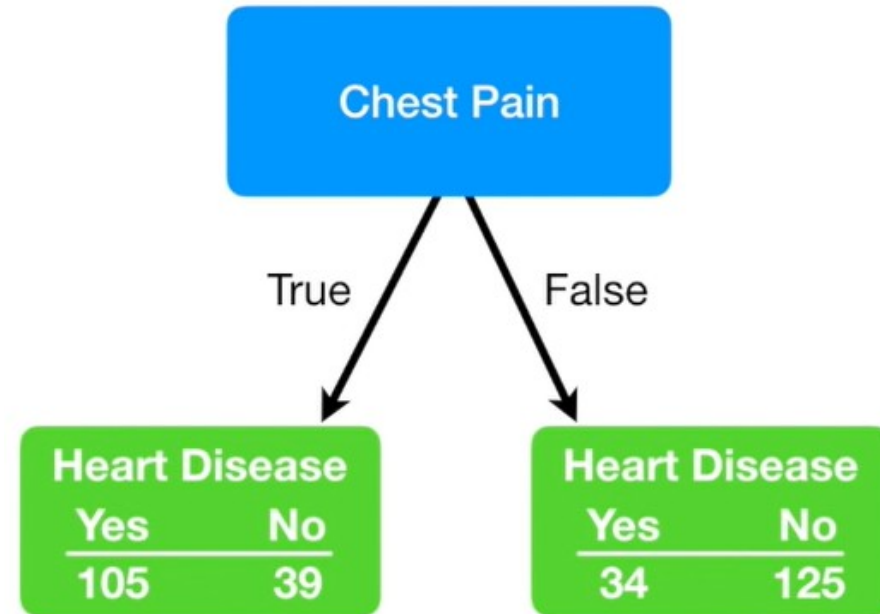


The image shows a cartoon of a man with a thoughtful expression, his hand on his chin, and a thought bubble above his head containing a question mark. To his right is a complex, branching tree diagram. The tree starts with a single purple node at the top, which branches into two orange nodes. These orange nodes further branch into a mix of orange, pink, and green nodes, creating a complex, non-linear structure that suggests a search space or a complex problem space.

How does it work?

- Comparing features one by one

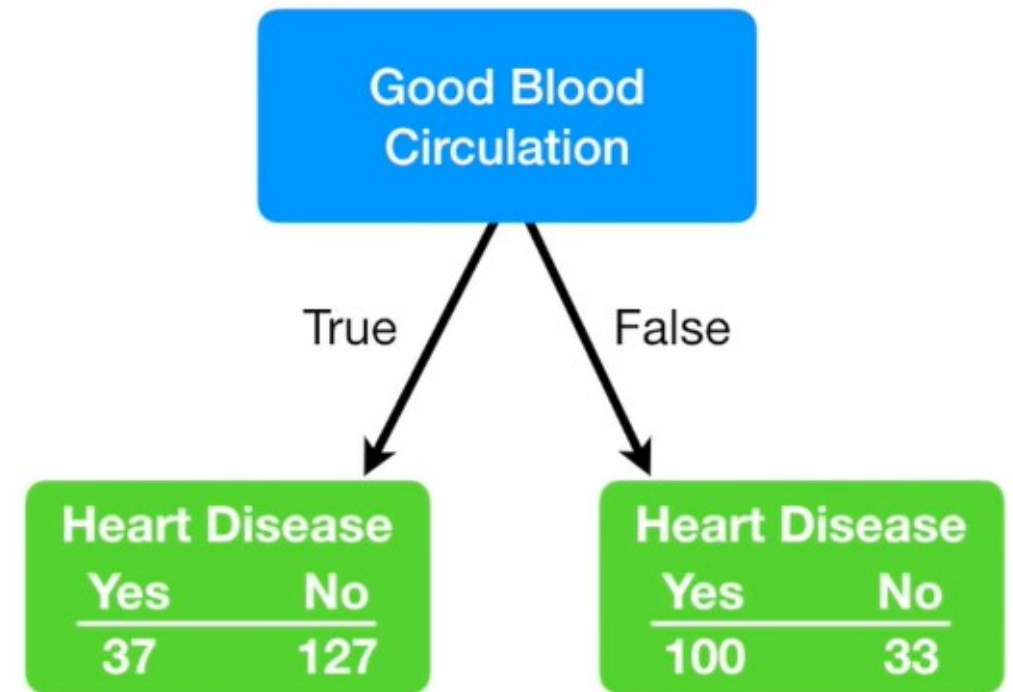
Chest Pain	Good Blood Circulation	Blocked Arteries	Heart Disease
No	No	No	No
Yes	Yes	Yes	Yes
Yes	Yes	No	No
Yes	No	???	Yes
etc...	etc...	etc...	etc...



How does it work?

- Comparing features one by one

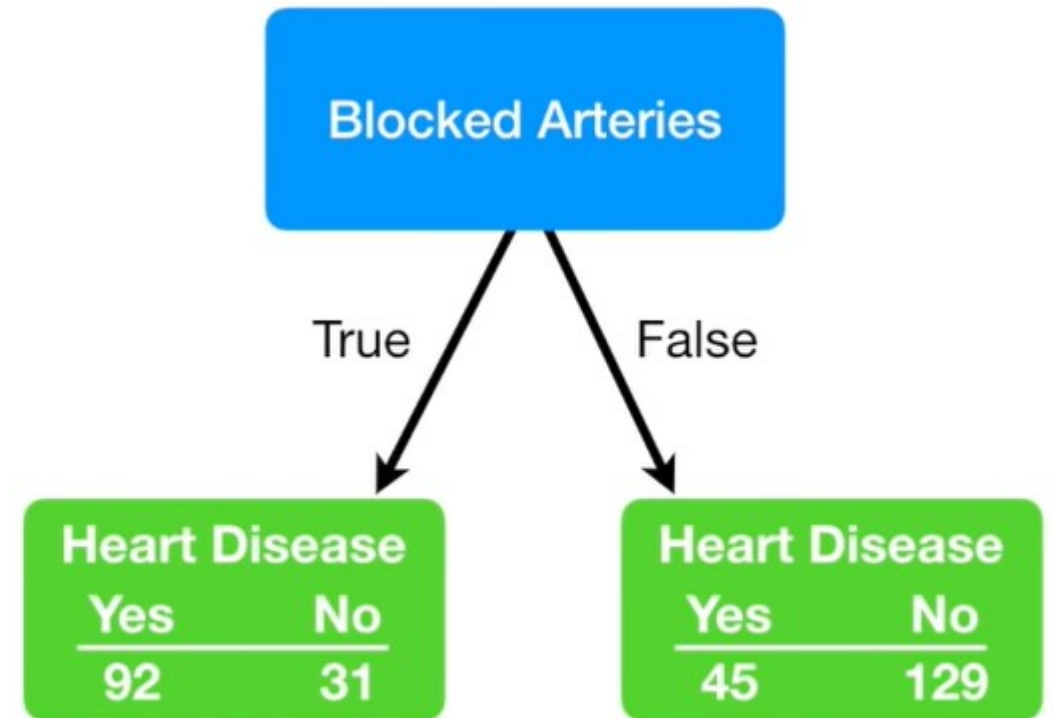
Chest Pain	Good Blood Circulation	Blocked Arteries	Heart Disease
No	No	No	No
Yes	Yes	Yes	Yes
Yes	Yes	No	No
Yes	No	???	Yes
etc...	etc...	etc...	etc...



How does it work?

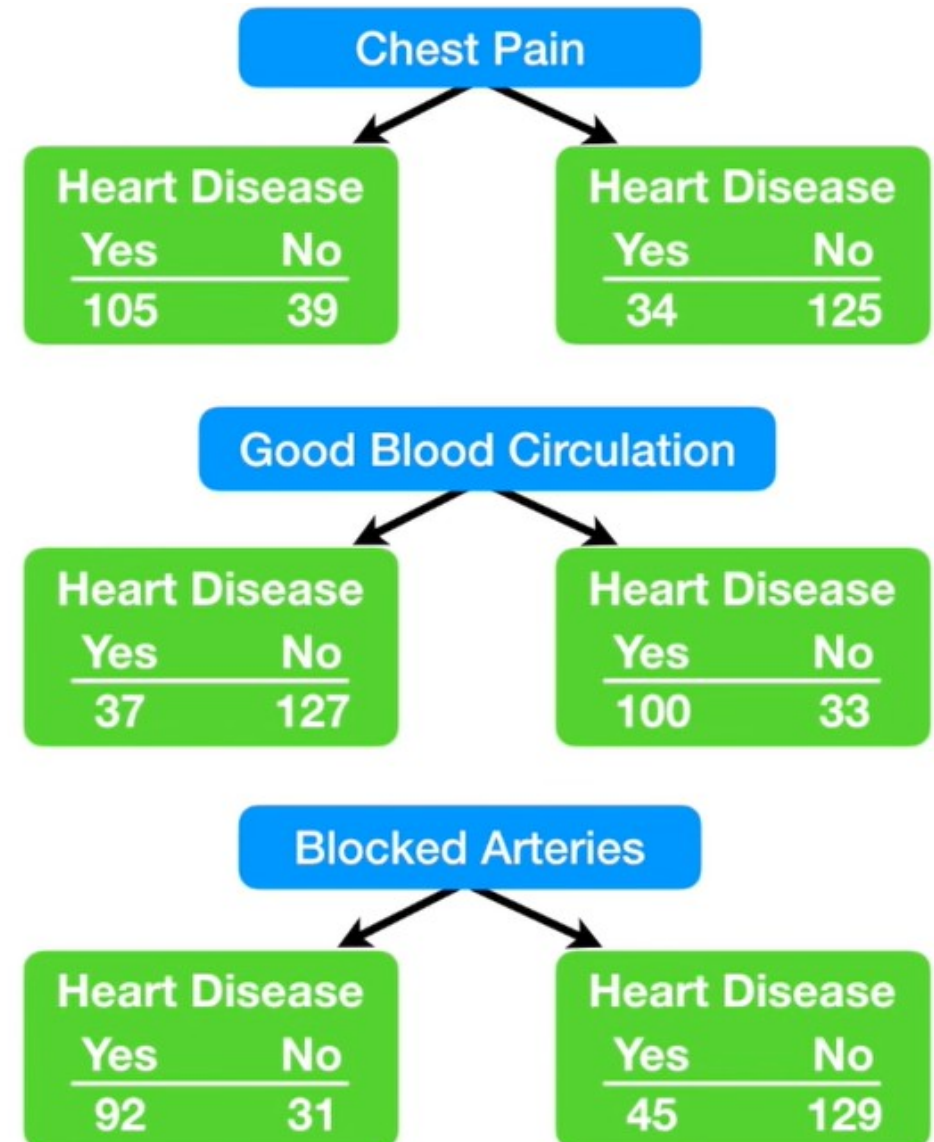
- Comparing features one by one

Chest Pain	Good Blood Circulation	Blocked Arteries	Heart Disease
No	No	No	No
Yes	Yes	Yes	Yes
Yes	Yes	No	No
Yes	No	???	Yes
etc...	etc...	etc...	etc...



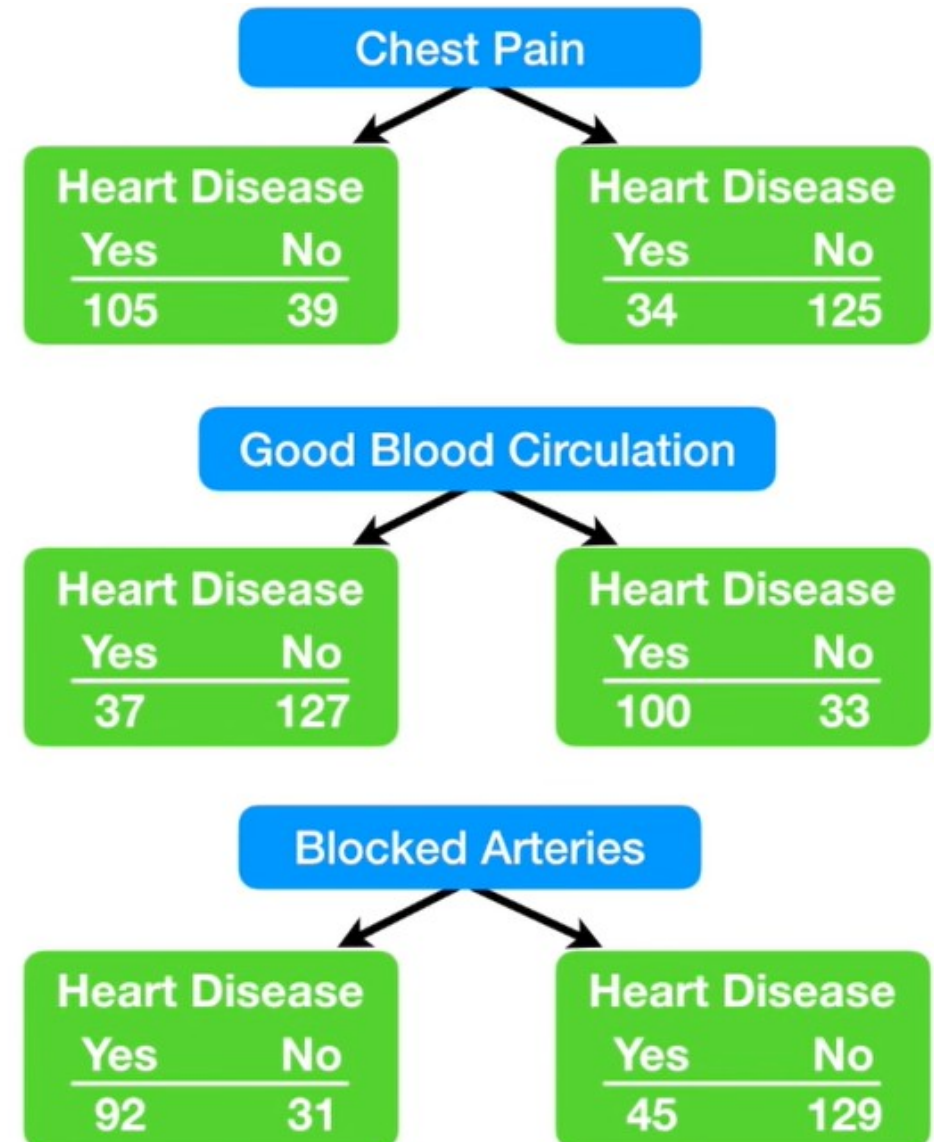
How does it work?

- No feature is perfect
- Total number of patients vary in each feature because of missing values



How does it work?

- Because none of the leaf nodes are 100% “Yes Heart Disease” or “No Heart Disease” they are “impure” (inaccurate)
- Now which feature’s classification is best here?
- To decide that we need a measure of impurity – Gini / entropy.



Fun fact

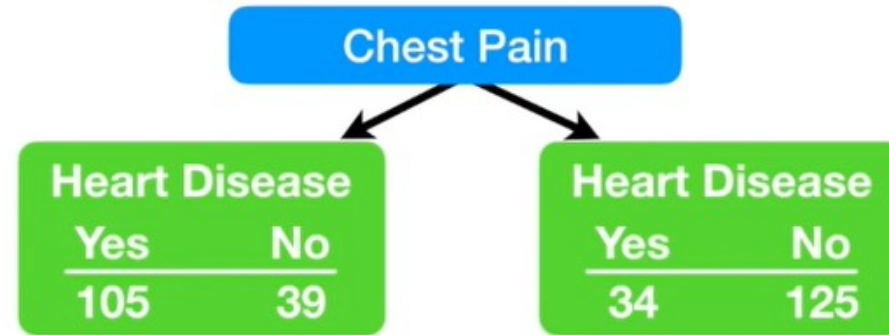
- Corrado Gini, inventor of the Gini coefficient, which measured income inequality.
- This got refined as anything that is unequal / explains unequally will be termed as impure.



Gini Vs. Entropy

- Gini measurement is the probability of a random sample being classified incorrectly if we randomly pick a label according to the distribution in a branch.
- Entropy is a measurement of information (or rather lack thereof). You calculate the information gain by making a split.

How does it work?

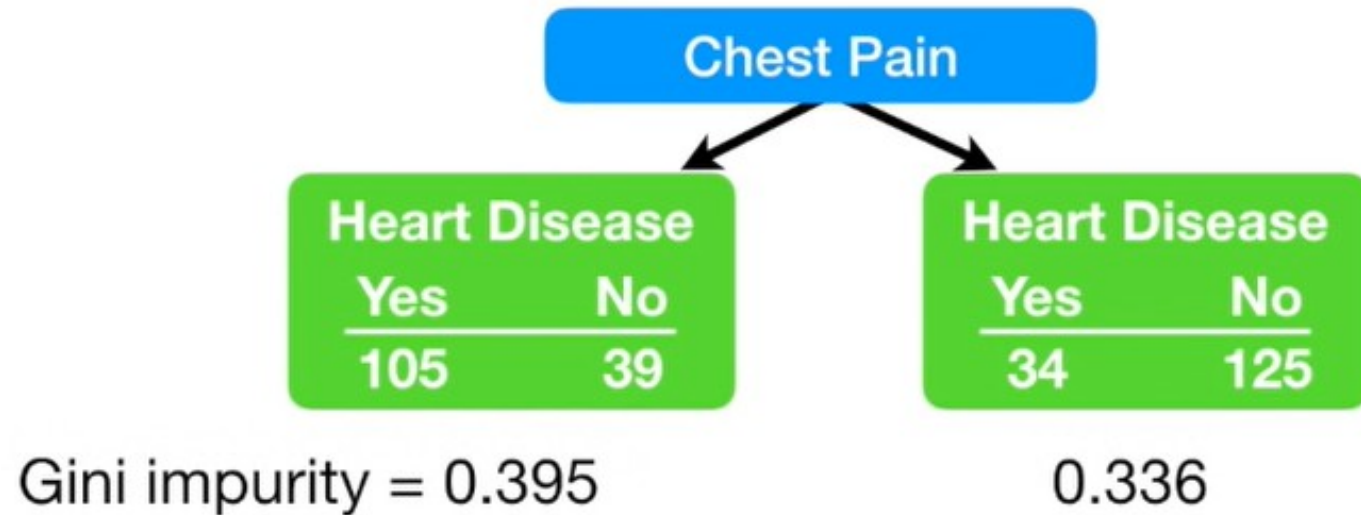


For this leaf, the Gini impurity = $1 - (\text{the probability of "yes"})^2 - (\text{the probability of "no"})^2$

$$= 1 - \left(\frac{105}{105 + 39}\right)^2 - \left(\frac{39}{105 + 39}\right)^2$$

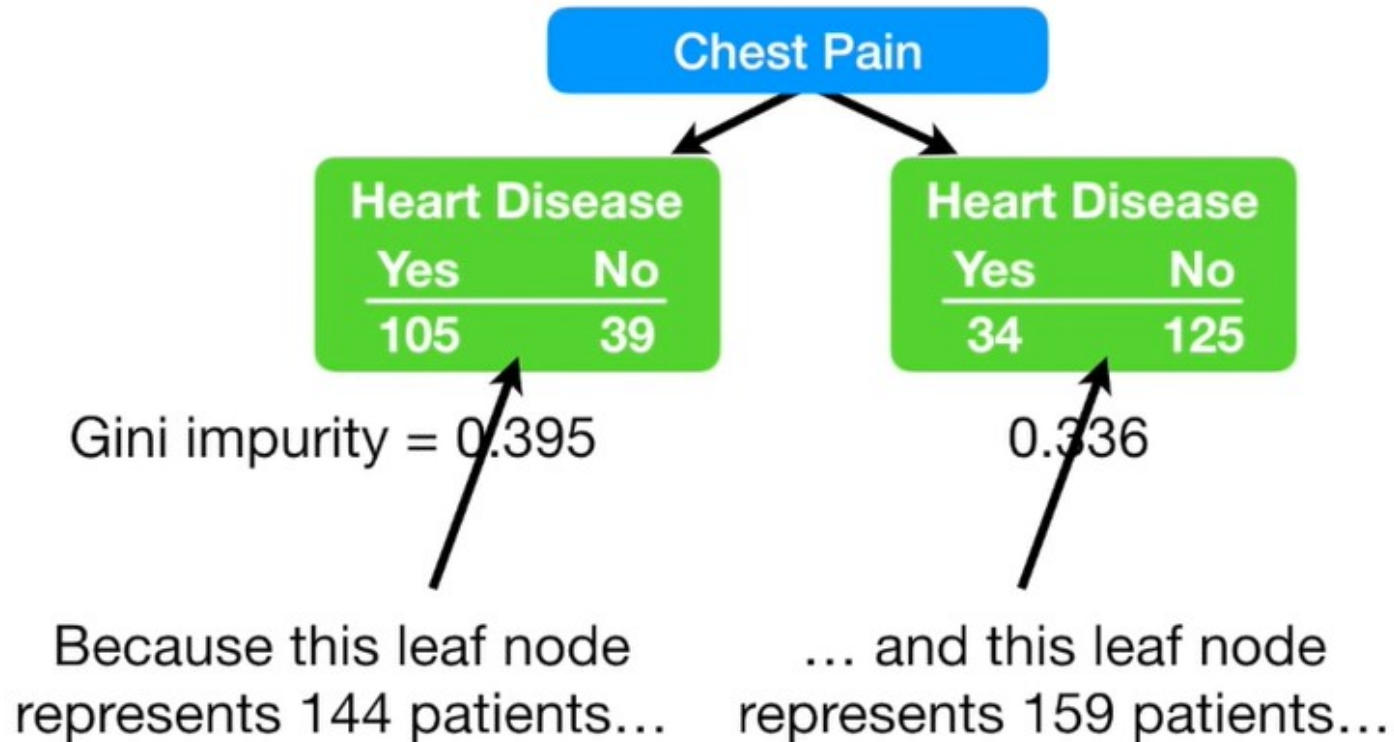
$$= 0.395$$

How does it work?



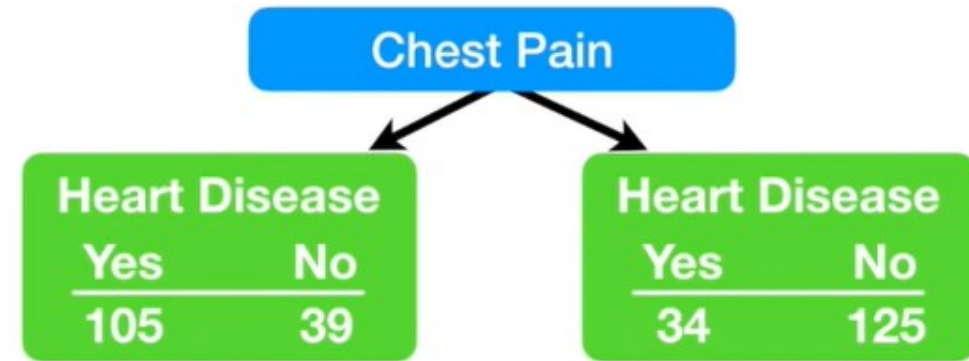
Now that we have measured the Gini impurity for both leaf nodes, we can calculate the total Gini impurity for using Chest Pain to separate patients with and without heart disease

How does it work?



Thus, the total Gini impurity for using Chest Pain to separate patients with and without heart disease is the **weighted average of the leaf node impurities**.

How does it work?



Gini impurity = 0.395

0.336

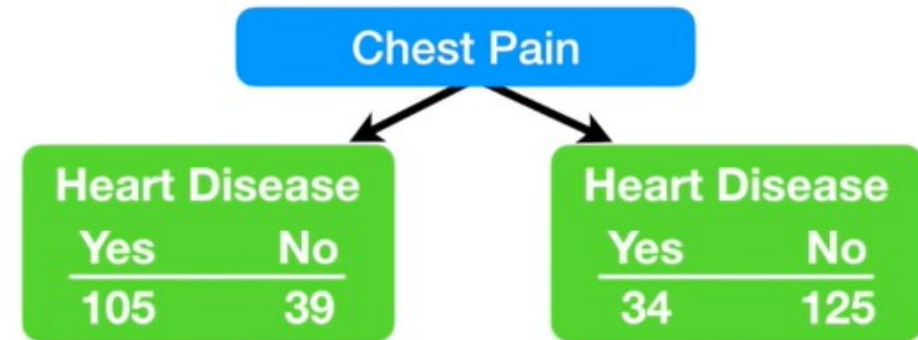
Gini impurity for Chest Pain = weighted average of Gini impurities for the leaf nodes

$$= \left(\frac{144}{144 + 159} \right) 0.395 + \left(\frac{159}{144 + 159} \right) 0.336$$

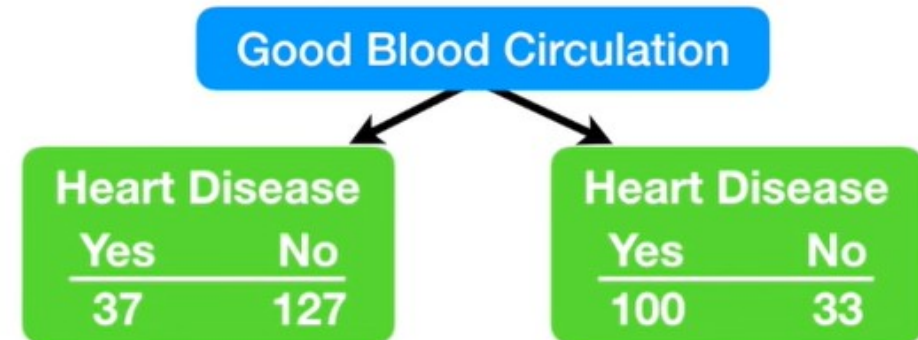
$$= 0.364$$

How does it work?

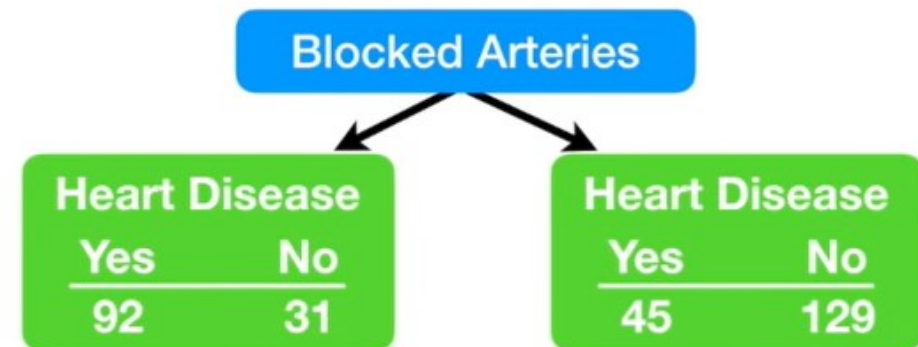
Gini impurity for Chest Pain = 0.364



Gini impurity for Good Blood Circulation = 0.360



Gini impurity for Blocked Arteries = 0.381



How does it work?

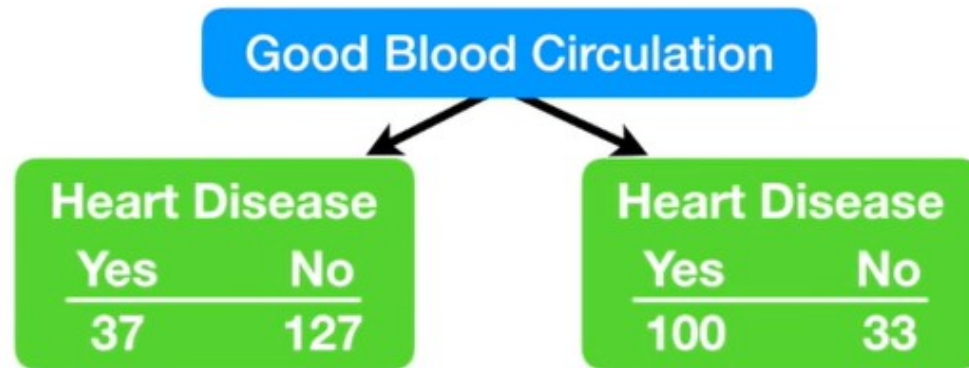
Gini impurity for Chest Pain = 0.364

Gini impurity for Good Blood
Circulation = 0.360

Good Blood Circulation has the lowest impurity (it separates patients with and without heart disease the best)...

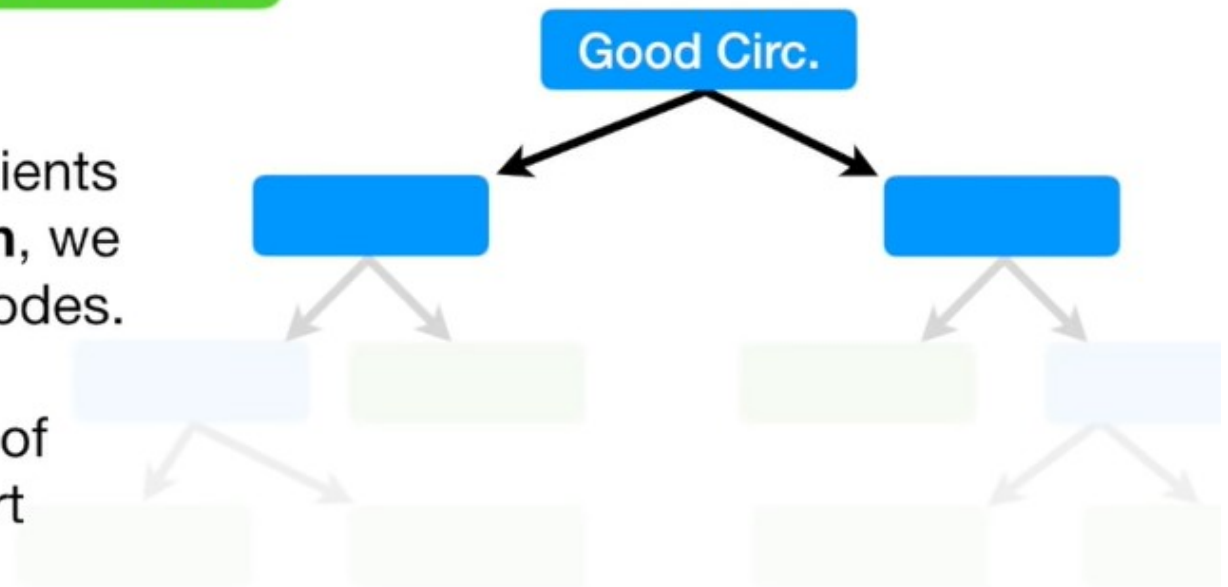
Gini impurity for Blocked Arteries = 0.381

How does it work?

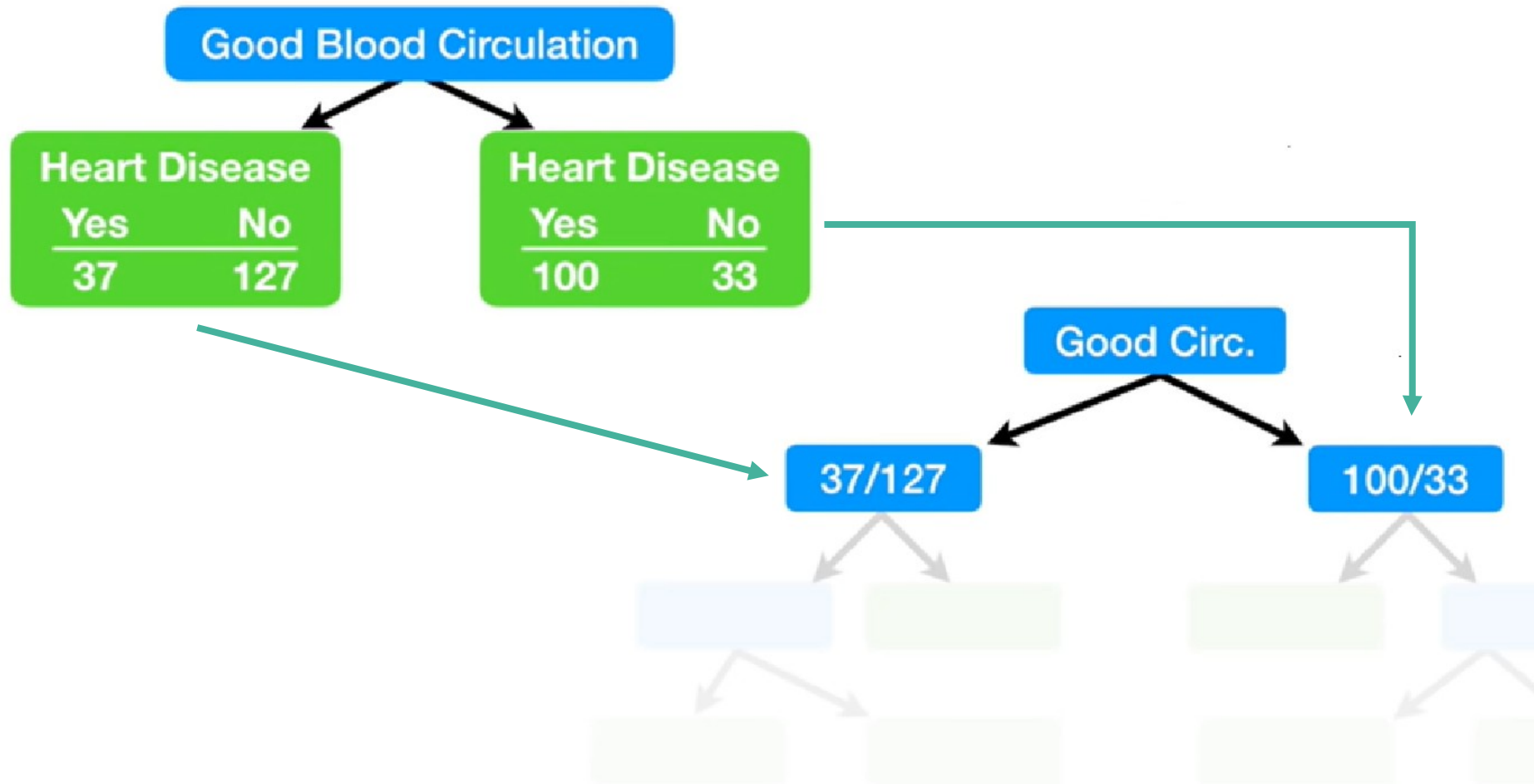


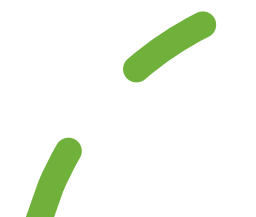
When we divided all of the patients using **Good Blood Circulation**, we ended up with “impure” leaf nodes.

Each leaf contained a mixture of patients with and without Heart Disease.



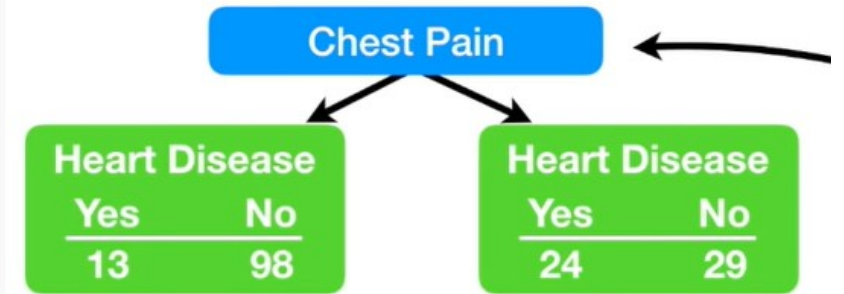
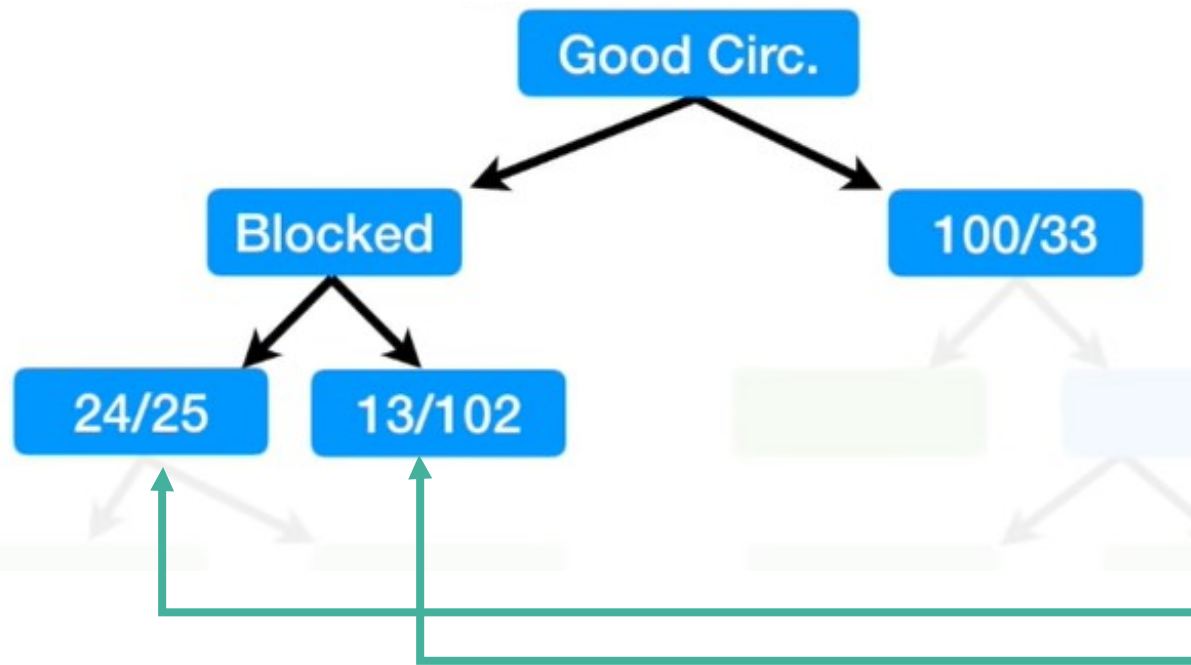
How does it work?



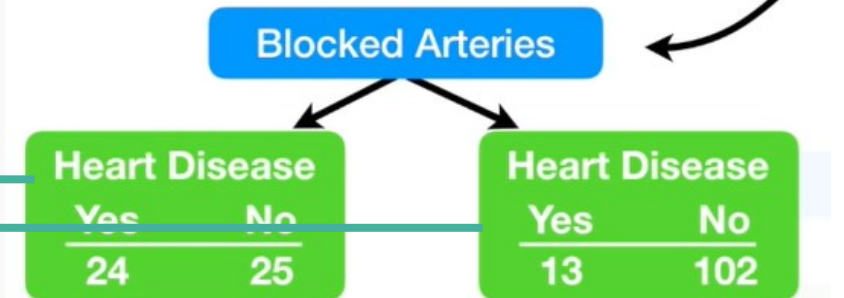


How does it work?

- Now the second internal node is “Blocked”



Gini impurity for Chest Pain = 0.3

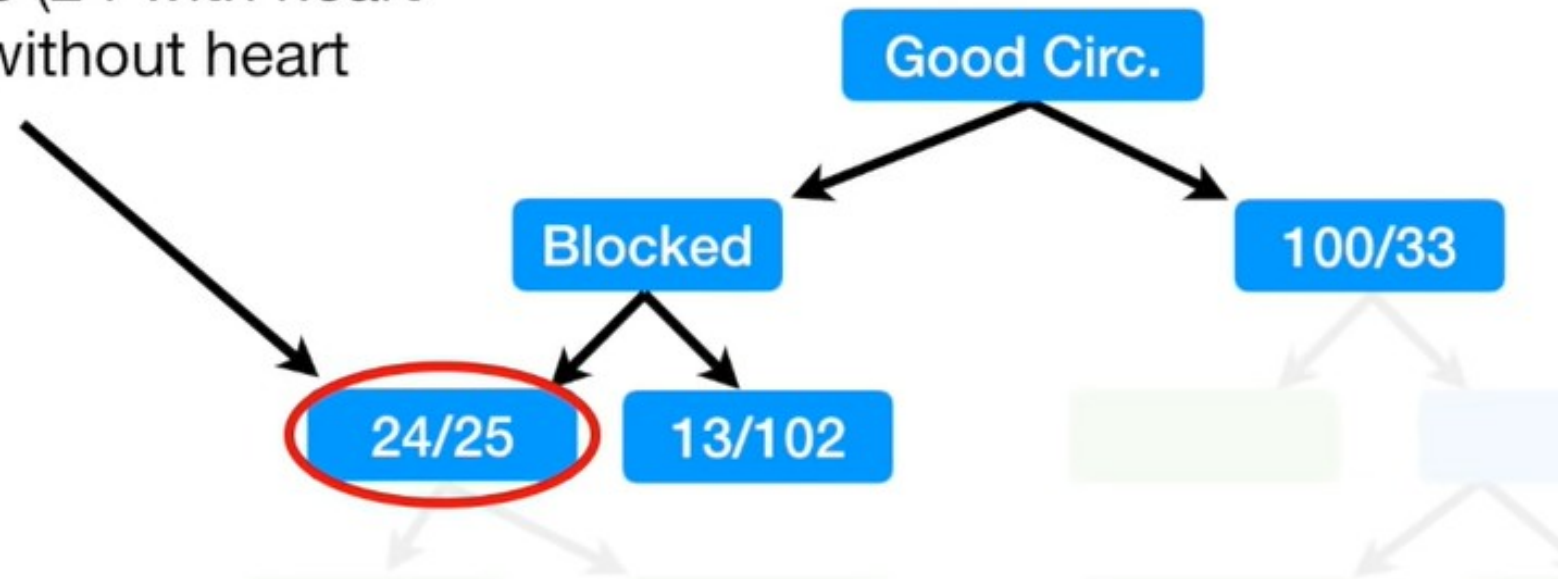


Gini impurity for Blocked Arteries = 0.2

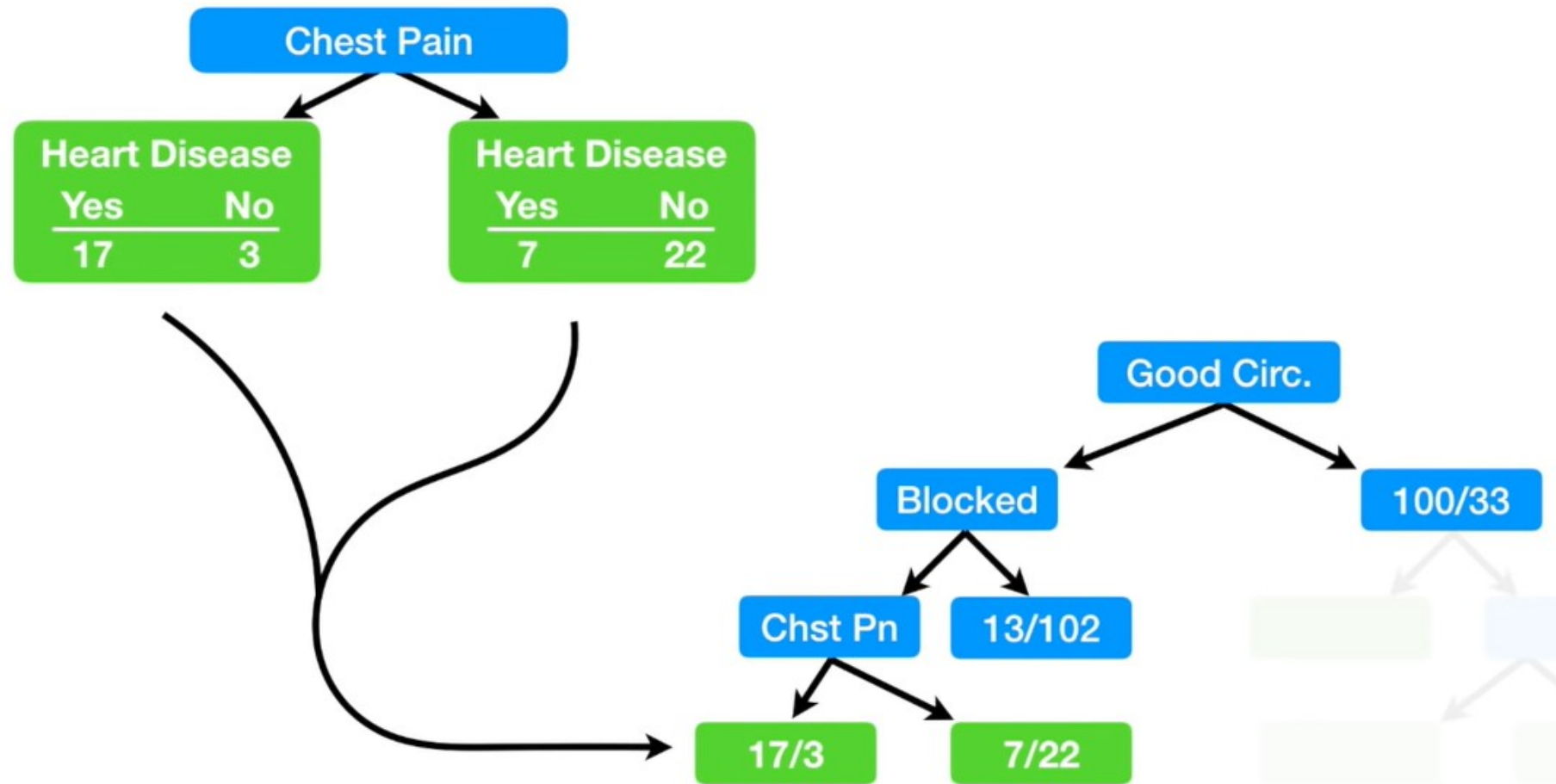
- Now we only have “Chest Pain” left

How does it work?

All we have left is Chest Pain, so first we'll see how well it separates these 49 patients (24 with heart disease and 25 without heart disease).



How does it work?

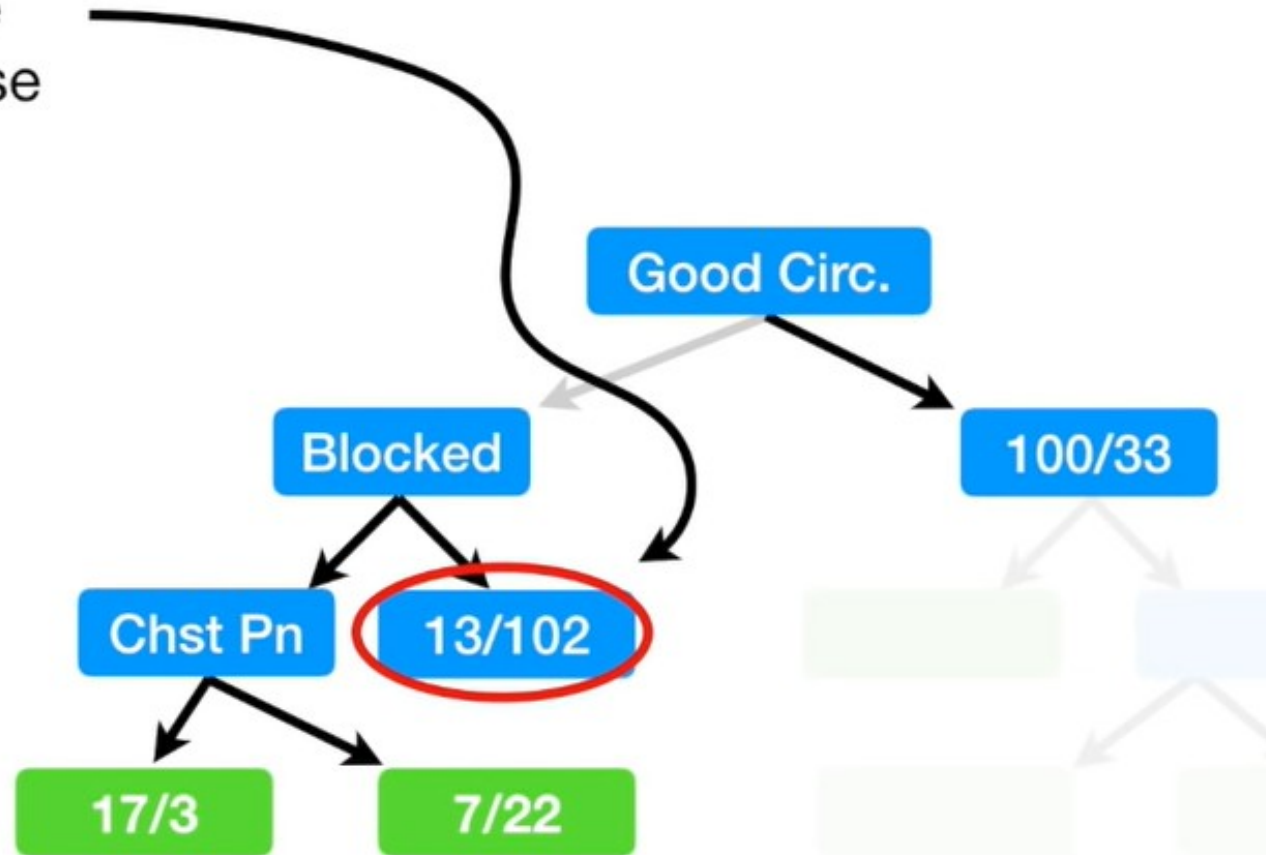


...so these are the final leaf nodes on this branch of the tree.

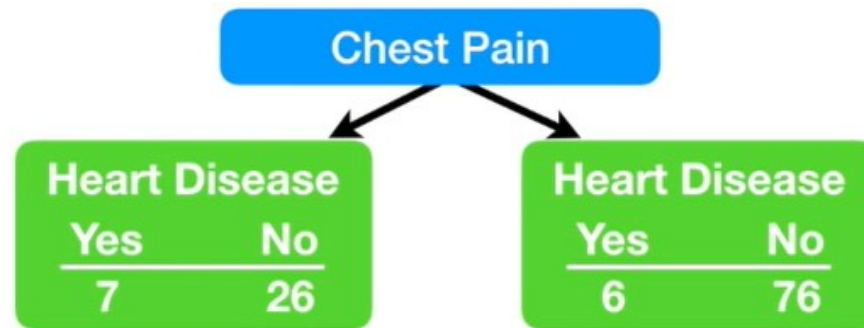
How does it work?

Now let's see what happens when we use chest pain to divide these 115 patients (13 with heart disease and 102 without).

NOTE: The vast majority of the patients in this node (89%) don't have heart disease.



How does it work?



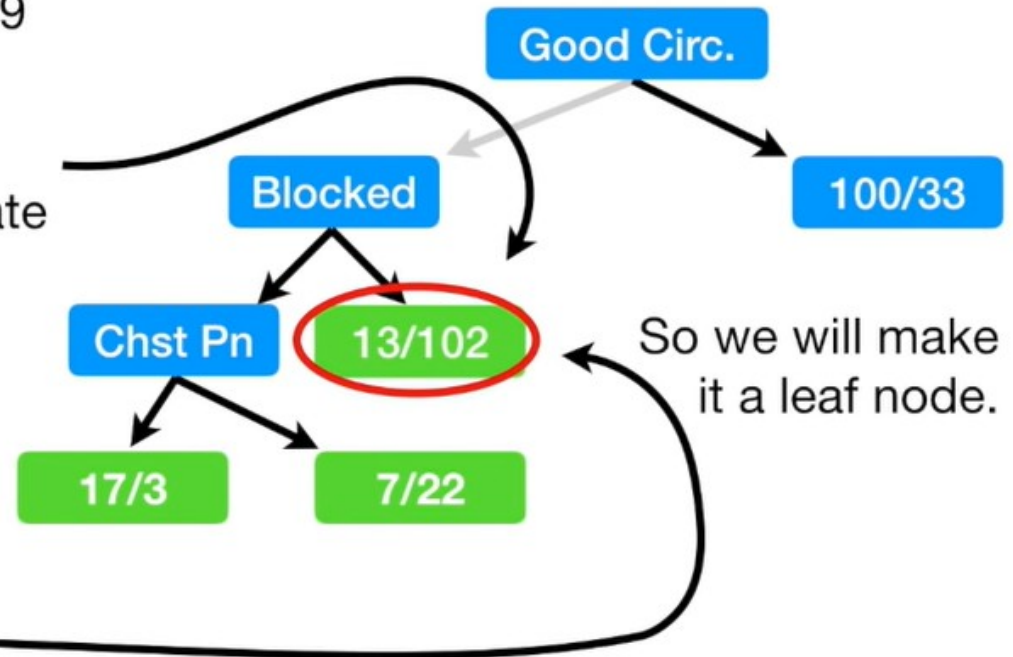
Gini impurity for Chest Pain = 0.29

The Gini impurity for this node, before using chest pain to separate patients is...

$$= 1 - (\text{the probability of "yes"})^2 - (\text{the probability of "no"})^2$$

$$= 1 - \left(\frac{13}{13 + 102}\right)^2 - \left(\frac{102}{13 + 102}\right)^2$$

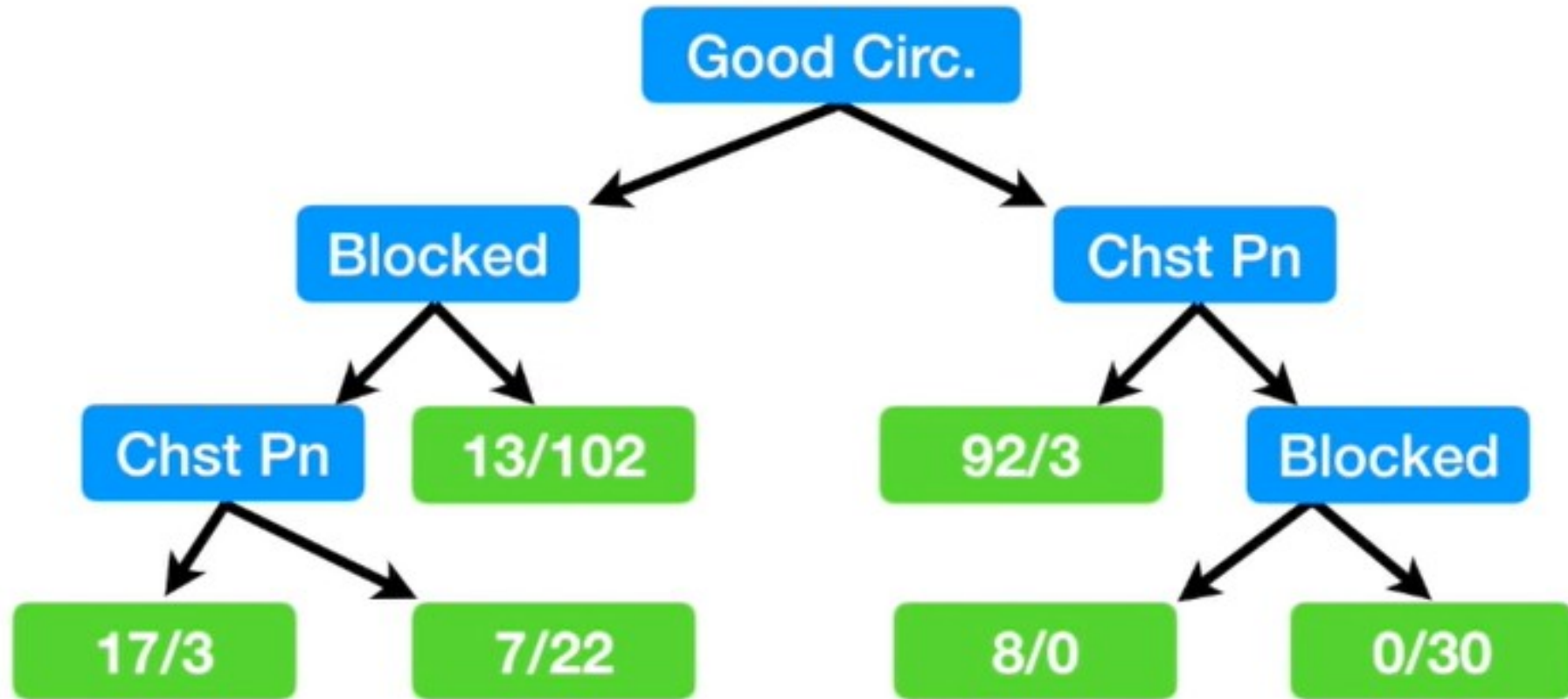
$$= 0.2$$



How does it work?

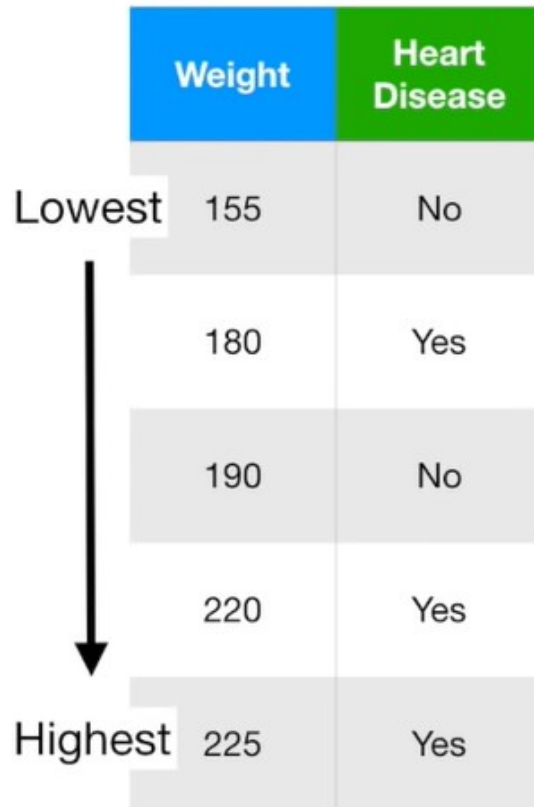
- The same rules to be followed on the right side of the tree
- Calculate all the Gini impurity scores
- If the node itself has the lowest score, then there is no point in separating the patients any more and it becomes the leaf node
- If separating the data results in an improvement then pick the separation with the lowest impurity value

How does it work?



How does it work?

- For numerical data?



	Weight	Heart Disease
Lowest	155	No
	180	Yes
	190	No
	220	Yes
Highest	225	Yes

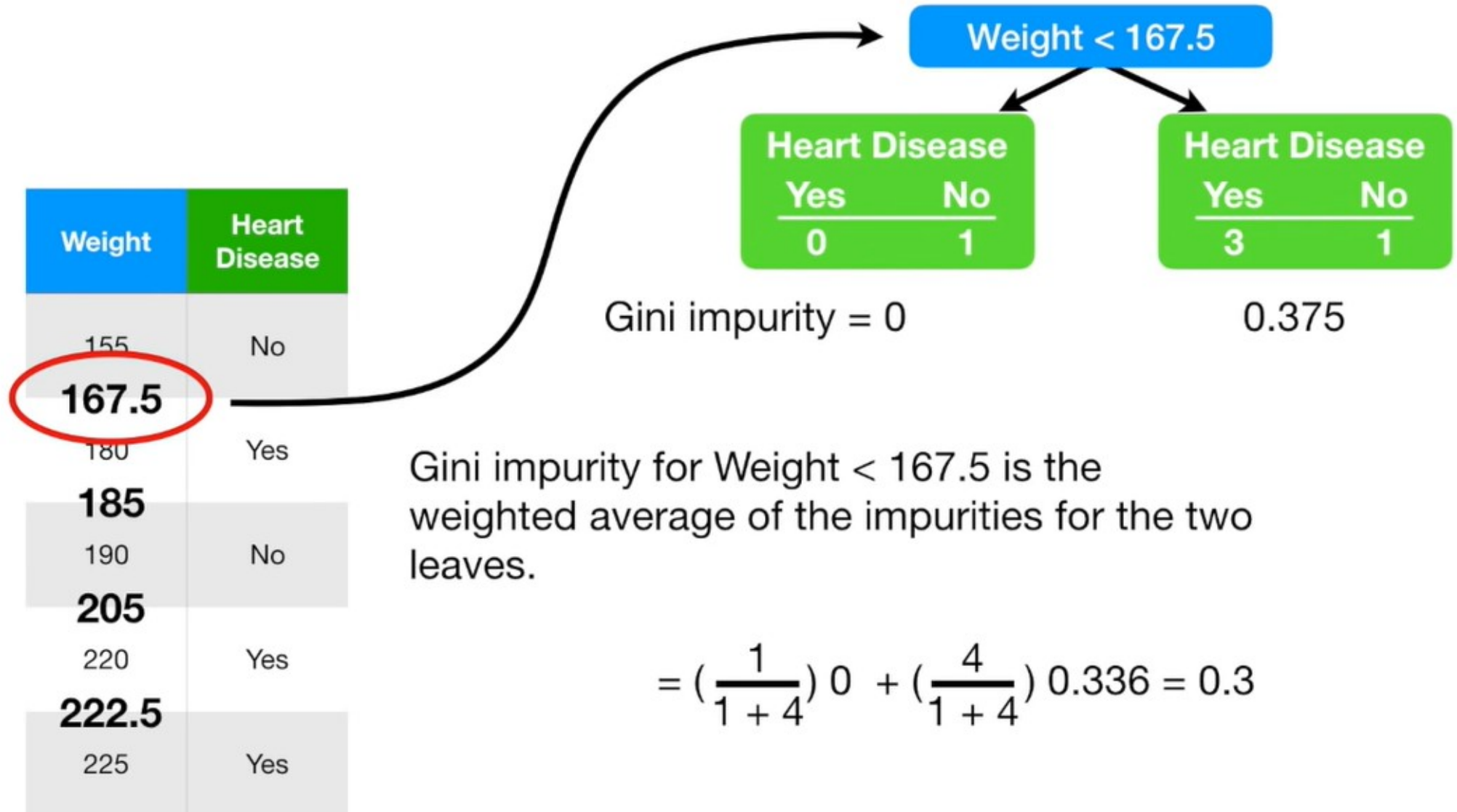
Step 1) Sort the patients by weight, lowest to highest.

How does it work?

Weight	Heart Disease
155	No
167.5	
180	Yes
185	
190	No
205	
220	Yes
222.5	
225	Yes

Step 2) Calculate the average weight for all adjacent patients.

How does it work?



How does it work?

Weight	Heart Disease
155	No
167.5	
180	Yes
185	
190	No
205	
220	Yes
222.5	
225	Yes

Gini impurity = 0.3

Gini impurity = 0.47

Gini impurity = 0.27

Gini impurity = 0.4

The lowest impurity occurs when we separate using **weight < 205...**

And follow the same rules as previous example

Many algorithms

- Decision Trees
- Renamed as CART
- ID3
- ID4.5
- C5.0
- Etc.

When to stop splitting?

- Many features usually mean many splits and a huge tree, which takes longer to build and can overfit: great accuracy on training data, poor accuracy on test data.
- One fix: limit the `max_depth` parameter. A higher `max_depth` allows a more complex (and more overfit-prone) tree.
- Training error keeps dropping as `max_depth` grows, but test accuracy gets worse past a point.
- Use `GridSearchCV` to find the `max_depth` that avoids both overfitting and underfitting.

When to stop splitting?

- `min_samples_split`: the minimum number of samples a node must have before it's allowed to split. Example: with a minimum of 10, a node with fewer than 10 samples stops splitting and becomes a leaf.
- `min_samples_leaf`: the minimum number of samples required in a leaf node.
- `max_features`: the number of features to consider when looking for the best split.

Pruning?

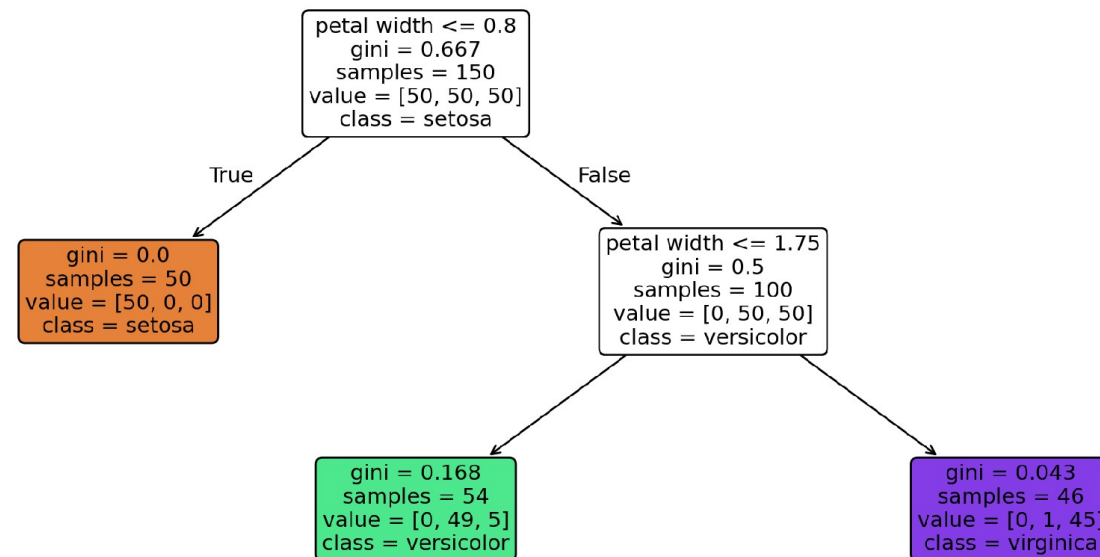
- Pruning helps avoid overfitting by cutting nodes or branches that add little value to the tree.
- Pre-pruning: stop growing the tree early, removing a node during construction if it looks unimportant.
- Post-pruning: grow the full tree first, then remove nodes afterward based on their significance.

What if?

- What if instead of 1 tree there was a way that multiple trees could work in parallel?
- Would that not be efficient? Especially with larger data size?
- Is there a way where imbalance does not impact my model?
- Faster computation perhaps?

Reading a Plotted Tree

- **Split rule:** the feature/threshold this node tests
- **gini / samples:** impurity, and how many rows reached this node
- **value / class:** label counts here, and the predicted class
- **Color intensity:** darker means purer (closer to one class)



Resources / References

- <https://mychartguide.com/decision-tree/>
- <https://statquest.com>
- <https://aionlinecourse.com>